

Google Merchandise Store: Production SQL Scripts

Complete repository technical archive containing only optimized BigQuery queries

1. Event Parameter Extraction (UNNEST Structure)

```
SELECT
  event_name,
  -- Flattening event parameters to extract target page locations
  (SELECT value.string_value FROM UNNEST(event_params) WHERE key = 'page_location') AS
  page_location,
  COUNT(DISTINCT user_pseudo_id) AS total_users,
  COUNT(*) AS total_events
FROM
  `bigquery-public-data.ga4_obfuscated_sample_ecommerce.events_*`
WHERE
  _TABLE_SUFFIX BETWEEN '20201101' AND '20201105'
  AND event_name IN ('page_view', 'view_item', 'add_to_cart', 'begin_checkout', 'purchase')
GROUP BY
  1, 2
ORDER BY
  total_events DESC
LIMIT 20;
```

2. Product Conversion Funnel (Option 1 — Gross Totals)

```
SELECT
  event_name,
  COUNT(DISTINCT user_pseudo_id) AS total_users,
  COUNT(*) AS total_events
FROM
  `bigquery-public-data.ga4_obfuscated_sample_ecommerce.events_*`
WHERE
  _TABLE_SUFFIX BETWEEN '20201101' AND '20201105'
  AND event_name IN ('page_view', 'view_item', 'add_to_cart', 'begin_checkout', 'purchase')
GROUP BY
  1
ORDER BY
  total_events DESC;
```

3. Strict Linear Product Conversion Funnel (Option 2 — Looker Studio Operational Core)

```
WITH funnel_steps AS (  
  SELECT  
    user_pseudo_id,  
    MAX(CASE WHEN event_name = 'session_start' THEN 1 ELSE 0 END) AS step_1,  
    MAX(CASE WHEN event_name = 'view_item' THEN 1 ELSE 0 END) AS step_2,  
    MAX(CASE WHEN event_name = 'add_to_cart' THEN 1 ELSE 0 END) AS step_3,  
    MAX(CASE WHEN event_name = 'begin_checkout' THEN 1 ELSE 0 END) AS step_4,  
    MAX(CASE WHEN event_name = 'purchase' THEN 1 ELSE 0 END) AS step_5  
  FROM  
    `bigquery-public-data.ga4_obfuscated_sample_ecommerce.events_*`  
  WHERE  
    _TABLE_SUFFIX BETWEEN '20201101' AND '20201130'  
  GROUP BY 1  
)  
  
SELECT '1. Session Start' AS stage, COUNT(*) AS users FROM funnel_steps WHERE step_1 = 1  
UNION ALL  
SELECT '2. View Item' AS stage, COUNT(*) AS users FROM funnel_steps WHERE step_1 = 1 AND step_2  
= 1  
UNION ALL  
SELECT '3. Add to Cart' AS stage, COUNT(*) AS users FROM funnel_steps WHERE step_1 = 1 AND  
step_2 = 1 AND step_3 = 1  
UNION ALL  
SELECT '4. Begin Checkout' AS stage, COUNT(*) AS users FROM funnel_steps WHERE step_1 = 1 AND  
step_2 = 1 AND step_3 = 1 AND step_4 = 1  
UNION ALL  
SELECT '5. Purchase' AS stage, COUNT(*) AS users FROM funnel_steps WHERE step_1 = 1 AND step_2  
= 1 AND step_3 = 1 AND step_4 = 1 AND step_5 = 1;
```

4. Gross Revenue & Channel Acquisition Monetization Model

```
WITH user_revenue AS (  
  SELECT  
    user_pseudo_id,  
    traffic_source.source AS traffic_source,  
    SUM(ecommerce.purchase_revenue_in_usd) AS user_lifetime_revenue,  
    COUNT(CASE WHEN event_name = 'purchase' THEN 1 END) AS total_purchases  
  FROM  
    `bigquery-public-data.ga4_obfuscated_sample_ecommerce.events_*`  
  WHERE  
    _TABLE_SUFFIX BETWEEN '20201101' AND '20201130'  
  GROUP BY  
    1, 2
```

```

),

source_aggregates AS (
  SELECT
    traffic_source,
    COUNT(DISTINCT user_pseudo_id) AS total_users_acquired,
    COUNT(CASE WHEN total_purchases > 0 THEN 1 END) AS paying_users,
    SUM(user_lifetime_revenue) AS total_source_revenue
  FROM
    user_revenue
  GROUP BY
    1
)

SELECT
  CASE
    WHEN traffic_source IS NULL THEN 'Organic/Direct'
    ELSE traffic_source
  END AS marketing_source,
  total_users_acquired AS users_acquired,
  paying_users,
  ROUND(SAFE_DIVIDE(paying_users, total_users_acquired) * 100, 2) AS conversion_to_buyer_rate,
  ROUND(total_source_revenue, 2) AS total_revenue,
  ROUND(SAFE_DIVIDE(total_source_revenue, total_users_acquired), 2) AS LTV_per_user
FROM
  source_aggregates
WHERE
  total_users_acquired > 100
ORDER BY
  total_revenue DESC;

```

5. Marketing Interaction Matrix (First-Touch vs. Last-Touch Attribution)

```
WITH ordered_events AS (  
  SELECT  
    user_pseudo_id,  
    event_timestamp,  
    event_name,  
    traffic_source.source AS current_source,  
    traffic_source.medium AS current_medium,  
    ROW_NUMBER() OVER (PARTITION BY user_pseudo_id ORDER BY event_timestamp ASC) AS row_asc,  
    ROW_NUMBER() OVER (PARTITION BY user_pseudo_id ORDER BY event_timestamp DESC) AS row_desc  
  FROM  
    `bigquery-public-data.ga4_obfuscated_sample_ecommerce.events_*`  
  WHERE  
    _TABLE_SUFFIX BETWEEN '20201101' AND '20201130'  
)  
  
first_touch AS (  
  SELECT user_pseudo_id, current_source AS first_source FROM ordered_events WHERE row_asc = 1  
)  
  
last_touch AS (  
  SELECT user_pseudo_id, current_source AS last_source FROM ordered_events WHERE row_desc = 1  
)  
  
SELECT  
  f.first_source AS first_touch_source,  
  l.last_source AS last_touch_source,  
  COUNT(DISTINCT f.user_pseudo_id) AS total_users_tracked  
FROM  
  first_touch f  
JOIN  
  last_touch l ON f.user_pseudo_id = l.user_pseudo_id  
GROUP BY  
  1, 2  
ORDER BY  
  total_users_tracked DESC  
LIMIT 5;
```

6. Cohort Retention Indexing (Day 7 & Day 30 Temporal Boundaries)

```
WITH user_first_visit AS (  
  SELECT  
    user_pseudo_id,  
    MIN(PARSE_DATE('%Y%m%d', event_date)) AS cohort_date  
  FROM  
    `bigquery-public-data.ga4_obfuscated_sample_ecommerce.events_*`
```

```

WHERE
  _TABLE_SUFFIX BETWEEN '20201101' AND '20201130'
GROUP BY 1
),

user_activity AS (
  SELECT
    e.user_pseudo_id,
    f.cohort_date,
    DATE_DIFF(PARSE_DATE('%Y%m%d', e.event_date), f.cohort_date, DAY) AS days_since_first_visit
  FROM
    `bigquery-public-data.ga4_obfuscated_sample_ecommerce.events_*` e
  JOIN
    user_first_visit f ON e.user_pseudo_id = f.user_pseudo_id
  WHERE
    e._TABLE_SUFFIX BETWEEN '20201101' AND '20201231'
)

SELECT
  FORMAT_DATE('%Y-%m', cohort_date) AS cohort_month,
  COUNT(DISTINCT user_pseudo_id) AS cohort_size,
  COUNT(DISTINCT CASE WHEN days_since_first_visit BETWEEN 6 AND 8 THEN user_pseudo_id END) AS
returned_day_7,
  ROUND(COUNT(DISTINCT CASE WHEN days_since_first_visit BETWEEN 6 AND 8 THEN user_pseudo_id
END) / COUNT(DISTINCT user_pseudo_id) * 100, 2) AS retention_rate_day_7,
  COUNT(DISTINCT CASE WHEN days_since_first_visit BETWEEN 28 AND 32 THEN user_pseudo_id END) AS
returned_day_30,
  ROUND(COUNT(DISTINCT CASE WHEN days_since_first_visit BETWEEN 28 AND 32 THEN user_pseudo_id
END) / COUNT(DISTINCT user_pseudo_id) * 100, 2) AS retention_rate_day_30
FROM
  user_activity
GROUP BY 1;

```

7. Cross-Platform Technical Diagnostics (iOS vs. Android Segmentations)

```
WITH raw_events AS (  
  SELECT  
    user_pseudo_id,  
    (SELECT value.int_value FROM UNNEST(event_params) WHERE key = 'ga_session_id') AS  
session_id,  
    event_name,  
    device.operating_system AS device_os  
  FROM  
    `bigquery-public-data.ga4_obfuscated_sample_ecommerce.events_*`  
  WHERE  
    _TABLE_SUFFIX BETWEEN '20201101' AND '20201130'  
)  
  
funnel_stages AS (  
  SELECT  
    user_pseudo_id,  
    session_id,  
    device_os,  
    MAX(CASE WHEN event_name = 'session_start' THEN 1 ELSE 0 END) AS step_1_session_start,  
    MAX(CASE WHEN event_name = 'purchase' THEN 1 ELSE 0 END) AS step_5_purchase_complete  
  FROM  
    raw_events  
  GROUP BY  
    1, 2, 3  
)  
  
SELECT  
  CASE  
    WHEN device_os IS NULL OR device_os = '(not set)' THEN 'Web / Unknown'  
    ELSE device_os  
  END AS platform_os,  
  COUNT(DISTINCT session_id) AS total_sessions,  
  SUM(step_5_purchase_complete) AS successful_purchases,  
  ROUND(SAFE_DIVIDE(SUM(step_5_purchase_complete), COUNT(DISTINCT session_id)) * 100, 2) AS  
total_conversion_rate  
FROM  
  funnel_stages  
WHERE  
  device_os IN ('iOS', 'Android')  
GROUP BY  
  device_os  
ORDER BY  
  total_sessions DESC;
```